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SHONGJOG

It works when the internet doesn't.

An offline-first, Bangla, voice-first emergency companion for Bangladesh.
Gemma 4 runs entirely on the phone, over a corpus verified against named health authorities.

The tools that answer emergency questions all stop working at the moment the emergency starts.

Every year floods and cyclones displace millions of people across Bangladesh. Cellular data, broadband and grid power are the first things to go – for hours, sometimes days.

While the network holds

Chatbots, translation apps and web helplines answer.

Hotlines connect and alerts arrive by SMS.

Someone with a signal can look things up for you.

After it fails

Every cloud assistant goes dark at the same moment.

A leaflet or a broadcast SMS cannot answer *your* situation.

The window before responders arrive is exactly when first-aid and safe-water decisions save lives.

And the people most exposed are the least served

Official guidance is English or dense Bangla text – unreadable at speed by rural, elderly and low-literacy users under stress. Meanwhile dangerous folk myths persist and actively harm: cut a snakebite, drink untreated flood water, stop drinking when you have cholera.

Three failures, and the one design decision each of them forces.

1

They assume a live connection

Chatbots, translation apps and helplines are all network-bound. They fail together, at the worst moment, for the same reason.

So Shongjog

runs the model on the phone. Airplane mode is the design target, not a degraded mode.

2

Static guidance can't answer a person

A leaflet or a broadcast SMS says the same thing to everyone. It cannot take "my child has diarrhoea and the water is dirty" and return the next step.

So Shongjog

answers the specific question, by voice, in Bangla — and reads the answer back aloud.

3

A general LLM will invent medicine

An ungrounded model hallucinates dosages and diagnoses. In first aid that is not a quality problem, it is a safety problem.

So Shongjog

answers only from a verified corpus, and never diagnoses or prescribes.

A Flutter app for Android with a 2.47 GB language model living inside it.

You ask an emergency question in Bangla — by voice or text. The phone understands it, retrieves verified guidance from a corpus stored on the device, generates a grounded step-by-step answer with Gemma 4, and reads it back aloud. No server is involved at any point.

48

verified Bangla corpus chunks
across 22 topics

263

cyclone shelters bundled
with GPS ranking

948

automated tests
passing, 0 failing

~44K

lines of Dart
across 156 files

Fully offline

Chat, family disaster planner, emergency-kit generator, risk assessment, situation summary, shelter brief, triage wizard, 26 quick cards, shelter map, emergency directory.

Better when online

Live hazard feeds from GDACS, NASA EONET and USGS; Open-Meteo weather, marine surge and air quality; OSRM routing; Nominatim and Overpass search. Every one fails soft.

No network at all

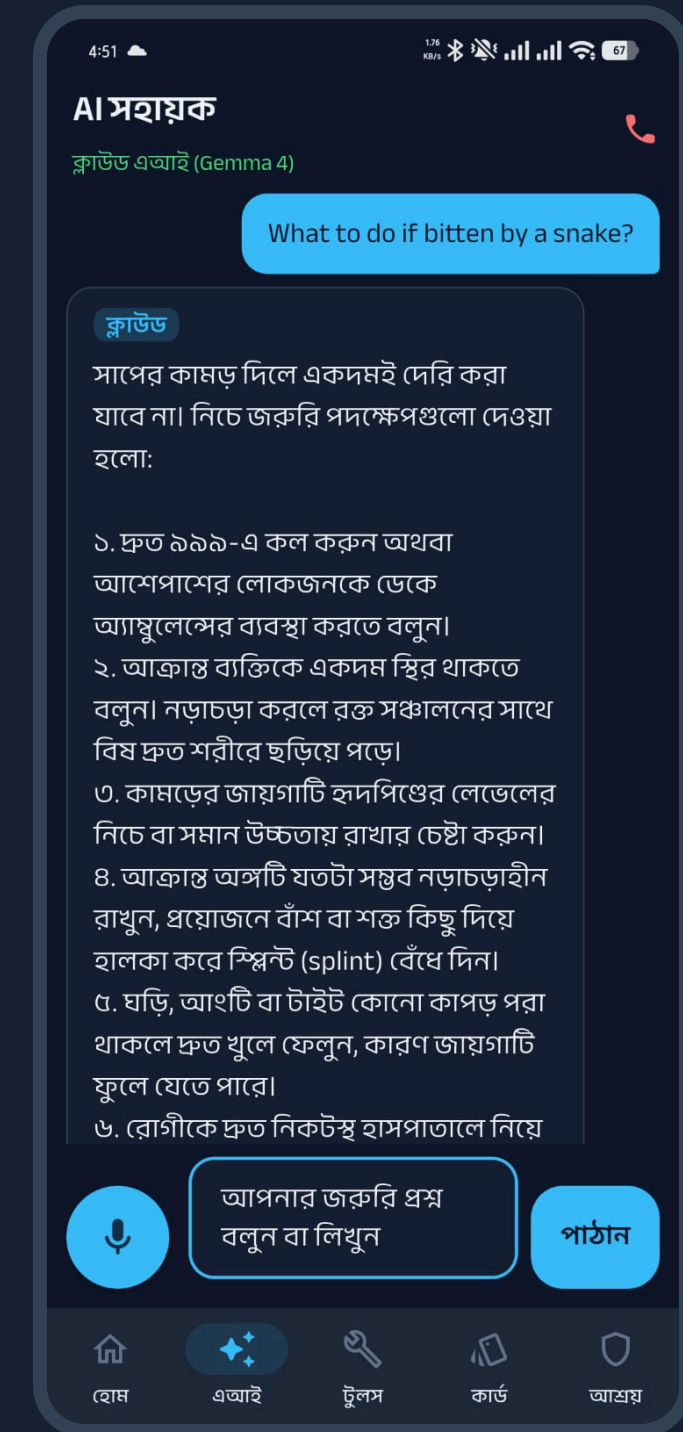
Phones form a mesh among themselves — text, photos, video, voice notes and full-duplex voice calls, plus a multi-hop SOS relay.

A real answer, generated on the phone, in airplane mode.

Asked in English — “*What to do if bitten by a snake?*” — and answered in Bangla, because that is the language the person in the emergency reads.

- Numbered, ordered steps — not a paragraph to parse under stress.
- Step 1 is ৯৯৯. The app routes to human help first, always.
- Grounded in the corpus chunk for snakebite, which cites WHO and BDRCS.
- It contradicts the folk myth directly: keep still, do not cut, do not tourniquet.
- A tap on পড়ুন reads it aloud for a user who cannot read it.

Bangla numerals throughout — ৯৯৯, not 999. Mixing digit systems inside one phrase is a defect in this product.



Four ways to answer, tried in order. The user always gets something.

Each tier is cheaper and more certain than the one above it. A failure at any level falls through silently rather than showing an error.

0

Shelter lookup — no model at all

"Where is the nearest shelter?" is answered by pure Dart: haversine distance over the 263 bundled shelters. It runs *before* every other tier — the fastest path serves the most time-critical question.

1

Gemma 4 E2B on the device

Retrieved corpus chunks and the question go to the local model through LiteRT-LM — the normal path, and the only one that works in airplane mode.

2

Cloud Gemini — only if reachable

If the device is online and the local model is still downloading or has failed, the same grounded prompt goes to Gemini, with key rotation and model fallback.

3

The corpus chunk, verbatim

No generation at all: the human-verified passage is shown as written. Less fluent, and impossible to hallucinate.

4

Call ৯৯৯

If nothing above produced an answer, the app says so plainly and offers the national emergency number. It never shows a blank bubble or a stack trace.

Why the shelter path has no model in it

It used to run a full inference just to extract one integer — how many shelters the user asked for — and on a miss it ran a *second* one. That was 30–50 seconds on mid-range hardware for the single most urgent question the app answers. It is now a parser and a distance sort: milliseconds, and it cannot hallucinate a shelter that does not exist.

Latency is a safety property

A classifier reads the question first. For a choking or bleeding query, extended reasoning is switched **off** — a correct answer in 40 s is worse than a good answer in 8 s. Generation is capped at 90 seconds with a fresh session per query, so a wedged model degrades into Tier 3 rather than hanging the screen.

No standard governs LLM-generated first aid. So we wrote the governance ourselves.

48

verified chunks

22

topics

34

cited documents

Every chunk is traceable to a named authority

A seven-organisation source whitelist: WHO, BDRCS, IFRC, CDC, UNICEF, MoDMR and BMD. Nothing enters the corpus from the open web, and every passage carries the document it came from — so a reviewer can check any answer against its origin.

Three rules the app cannot break

- It answers only from the corpus — it does not answer from the model's own memory.
- It never diagnoses and never prescribes a dose.
- It routes to ৯৯৯ or the nearest hospital for anything beyond first aid.

The 22 topics

ORS · diarrhoea · water · hygiene · fever · bleeding · CPR · drowning · snakebite · animal bite · respiratory · electrical · heat · cold · cyclone · food · infant · pregnancy · elderly · livestock · emotional support · documents

Correcting myths is a first-class job

Ten of the fifty evaluation queries are myths phrased as questions — "you should cut a snakebite, right?" The corpus contains the correction, and the answer contradicts the premise instead of politely working around it.

Two models on the handset, and a hard constraint that shapes everything.

Generation

Gemma 4 E2B — 2.47 GB

Quantised `.litertlm`, executed by LiteRT-LM through `flutter_gemma`. 1024-token context, 256-token output ceiling, temperature 0.3.

A fresh session is opened per query and closed in a `finally` block. Sharing one session across queries leaked KV-cache memory until the app was killed by the OS mid-answer.

Retrieval

EmbeddingGemma 300M — 179 MB

A `.tflite` embedder with its SentencePiece tokenizer, for true semantic search over the corpus.

The bundled vectors were built offline with a different model, and a query vector from one model is not cosine-comparable with a document vector from another. So on first run the app re-embeds all 48 chunks with the same on-device embedder and caches the index; later launches load it in about a millisecond.

arm64-v8a only

The inference engine ships native libraries for arm64 alone. A universal APK would still install on other ABIs with no engine at all — and fail in a way that looks exactly like a bug. Restricting the ABI makes that impossible and drops about 36 MB.

The key is never in the binary

A `--dart-define` value is a plaintext literal in the shipped library. The cloud key is fetched at runtime from Firestore into the Android Keystore instead — out of the binary, and revocable without a new release.

Voice on both ends

Bangla speech-to-text prefers a fully offline Vosk model and falls back to the platform recogniser; Bangla text-to-speech targets bn-BD with a bn-IN fallback, with opt-in auto-read for users who cannot read the screen.

If the towers are down, the phones become the network.

Nearby handsets discover each other directly and carry messages between people who have no carrier, no Wi-Fi and no internet.

What it carries

Text, photos, video and voice notes — plus full-duplex 8 kHz voice calls between two phones with no cellular service at all.

How it connects

`nearby_connections` in `P2P_CLUSTER` mode, whose transport is Wi-Fi Direct. Bluetooth is used only to advertise and scan.

When Google services are missing

Many phones sold in this market ship without GMS, which Nearby requires. A GMS-free Wi-Fi Direct implementation takes over on those devices.

The SOS relay

A distress message is re-broadcast by every phone that receives it — an LRU dedup over 256 message ids, a 5-hop ceiling and a 1-hour lifetime, so it spreads outward without looping forever.

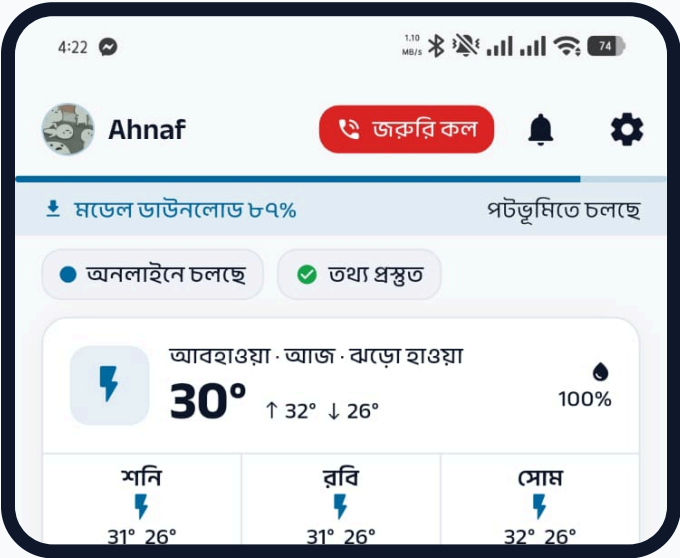
This is the part judges should ask us to break

We will hand you two phones, put both in airplane mode with Wi-Fi on, and send a message between them in front of you. Nothing in the room is involved — no venue Wi-Fi, no hotspot, no server.

An honest limit

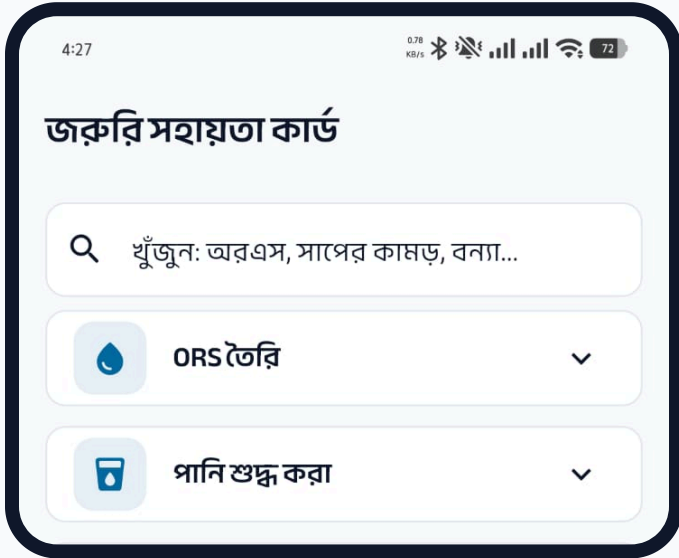
Range is line-of-sight Wi-Fi Direct — tens of metres, not kilometres. The mesh is for a shelter, a village cluster or a rescue team, not a district. We are not claiming otherwise.

Bangla first, everywhere – not an English app with a translation layer.



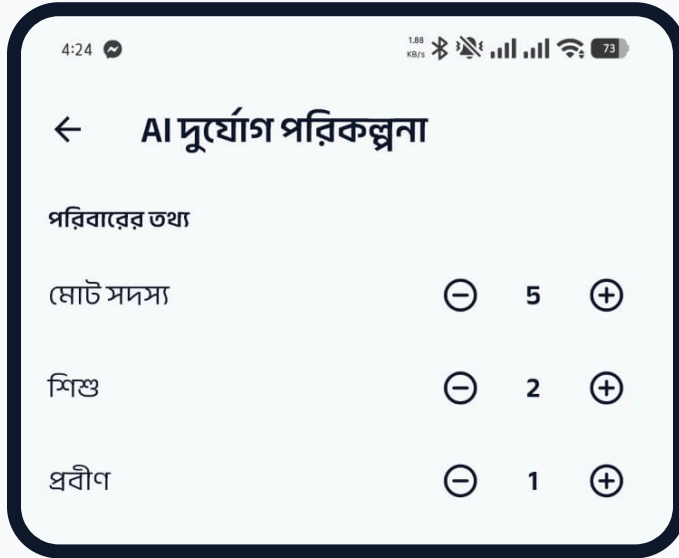
Home

Live weather, hazards and air quality when online; one tap to the triage wizard.



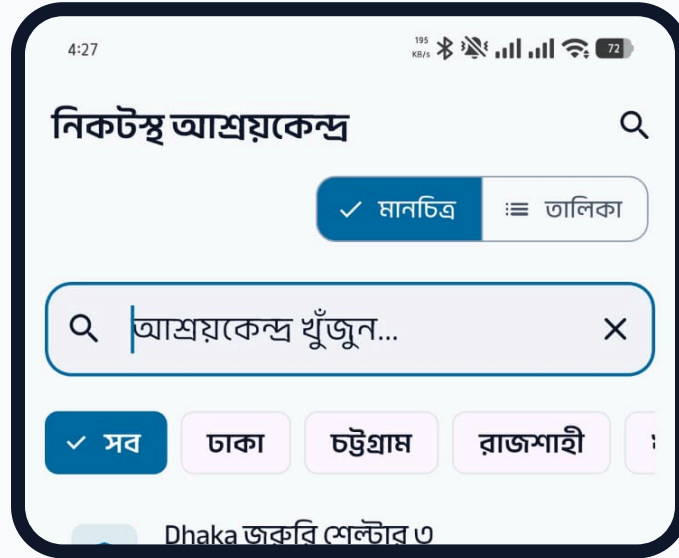
26 quick cards

Searchable, offline, no model needed. A red badge means life-threatening.



Family planner

Household details in, a plan for that family out – written by Gemma on the device.



Shelter finder

263 shelters ranked by GPS distance, with capacity and listing agency.

Voice-first by default

The microphone is the primary input, not a secondary one, and answers can be read aloud – because the user we designed for may not read fluently.

799 strings in each language

A complete English locale sits behind a toggle. Bangla is the default and the design language; English is the accommodation.

Type sized for stress

Body text floors at 17sp, not Material's 14sp, and tap targets at 48dp – the app is read outdoors, at night, by frightened people.

Survivors are not the only user. Someone has to coordinate them.

A coordinator panel, gated behind a login, shows the live picture: who has marked themselves safe, who has reported danger and where, which campaigns await approval, and how many devices are reachable.

Live safety roll

Counts sync through Firestore as people tap their status. Each danger row carries name, phone, type, time and a maps link.

Campaign approval

A resident requests a relief or rescue campaign; an approved one appears on every nearby user's map.

Broadcast

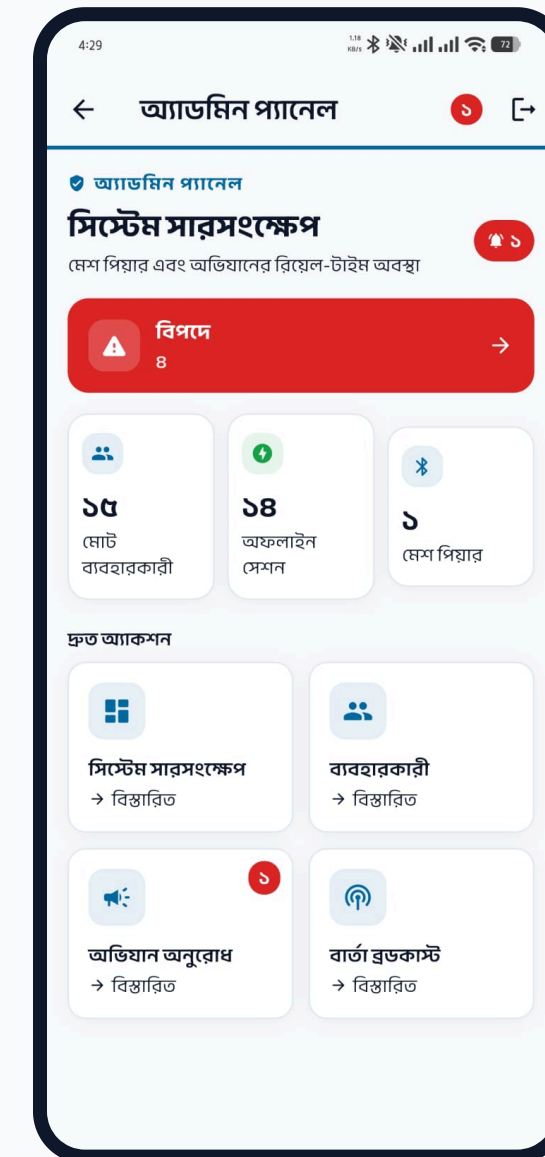
One message becomes a notification on every install – the highest-consequence control in the product.

Device roster

Every install that has ever had a connection, with last-seen time, and a marker for those reachable over the mesh right now.

We wrote down the risk we could not remove

Our Firestore rules file says in its own header that the admin gate is client-asserted and non-cryptographic, and that granting it unlocks a channel pushing notifications to every device – *"a misinformation channel, not merely a settings panel."* Naming it in the code is the honest answer available to a team with no trusted server.



One rule holds 156 files together: the parts that must be correct never touch the framework.

Presentation

features/ – 26 modules

chat · shelter · mesh_comm · triage · planner · emergency · voice · admin · hazards · weather · safe_beacon · damage_scanner · quick_cards · onboarding · settings ...

app/ – shell & theme

Routing, the five-tab shell, and one design system: colour tokens, a 17sp body floor, 12/16 dp radii, semantic tones.

Platform edge

core/ – services

model_manager · embedder_service · remote_key_service · device_registry · local_notification_service · sms_channel · firebase_auth · connectivity

Native & cloud

LiteRT-LM · Nearby Connections · Wi-Fi Direct · Firestore · 10 keyless public data services · Gemini.

Correctness core

rag/ · knowledge/

Retrieval, prompt construction, urgency classification, repetition detection, rumour checking, the triage decision tree, shelter ranking.

Pure Dart. Zero Flutter imports, zero plugin imports.

Because this layer imports nothing from the framework, the logic that decides what a person in an emergency is told can be tested exhaustively on a laptop, with no device and no emulator. That is why the suite can be this large and this fast.

Offline by construction

Every network call has a deterministic fallback. There is no code path where a dead network produces an error screen.

Deterministic where it counts

The triage wizard is a decision tree with eight terminal routes and no model in it. It cannot hallucinate.

Fails soft, loudly logged

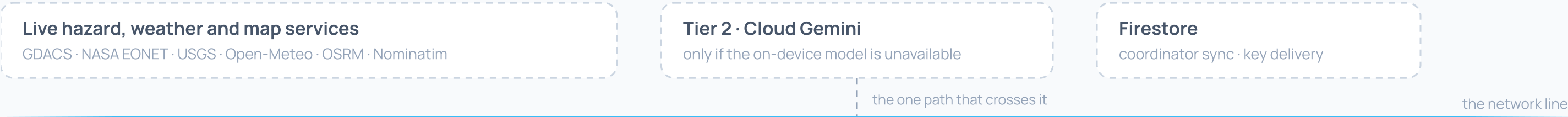
Ten public data services can each go down without the user seeing anything worse than a missing card.

Bilingual to the edges

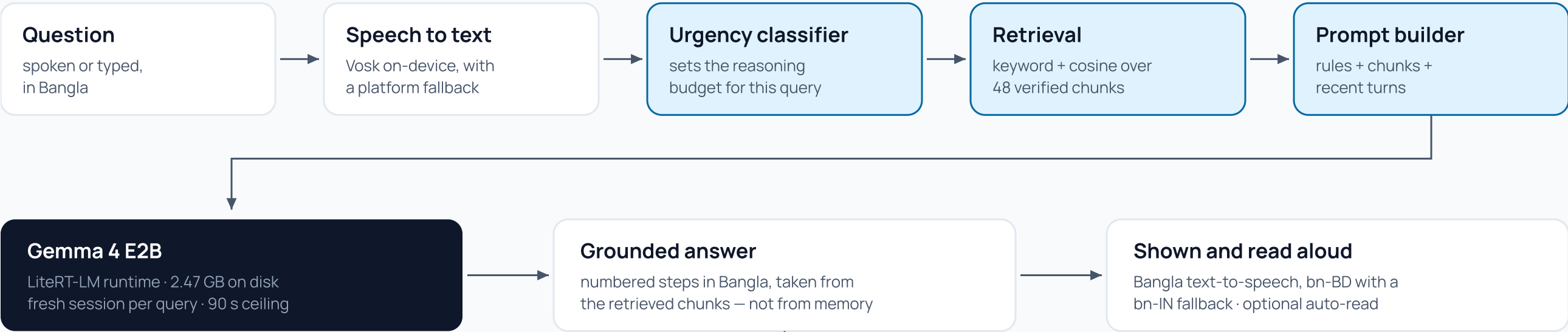
Numerals, relative times and error copy all localise — a Bangla screen never shows Latin digits.

One question, end to end – and exactly where the network line falls.

Off the device – reachable only when there is a network



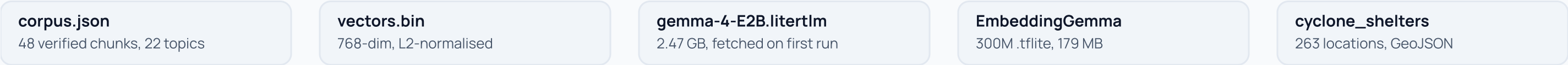
Everything below this line runs with the radio off



empty or failed → the verbatim chunk, then য্য্য

Pure Dart Platform Native

Read from local storage – shipped in the APK, or downloaded once



Everything on this slide was re-run today, on this branch.



The release gate exists because green tests were not enough

Both times the offline model broke, `flutter analyze`, `flutter test` and `flutter run` were all clean — the failure only appeared in an installed release APK. So the build script now inspects the artefact itself and refuses to ship if any check fails.

- The inference engine's native libraries are present for arm64.
- No other ABI is packaged, so Android cannot install an engineless build.
- No API key of any shape appears in the compiled Dart library.
- The notification icon survived resource shrinking.
- The `<queries>` intents for speech, camera and file-picking are declared — the same omission has silently broken four separate features on Android 11+.

Retrieval quality — measured, and not yet where we want it

A 50-query evaluation set across five categories, scored automatically.

Metric	Baseline	Current
Recall@1	46%	46%
Recall@3	60%	62%
Retrieval rate	98%	98%
Out-of-scope	0% by design — correctly retrieves nothing	

We are reporting the number that does not flatter us. Recall@1 is the weakest part of the system, the corpus tripled without it moving, and semantic retrieval with the on-device embedder is the work aimed squarely at it.

Three targets the app addresses directly, and three it supports.

Goal	Target	What in Shongjog serves it
11 · Sustainable Cities	11.5 — reduce deaths and the number of people affected by disasters	263 bundled cyclone shelters with offline GPS ranking and routing; multi-hop SOS relay; slide-to-confirm 999 dialer; SOS SMS with coordinates formatted for the operator.
13 · Climate Action	13.1 — strengthen resilience and adaptive capacity to climate hazards	Twelve hazard types; live GDACS, NASA EONET, USGS and Open-Meteo feeds; cyclone, flood, heat and cold guidance that keeps working after the feeds go dark.
3 · Good Health	3.d — strengthen capacity for early warning and risk management	48 chunks cited to WHO, BDRCS, IFRC, CDC and UNICEF; a triage wizard with eight terminal routes; ORS, CPR and bleeding guidance available with no network.
10 · Reduced Inequalities	10.2 — empower and include everyone, irrespective of status	Bangla-first and voice-first for low-literacy users; runs on a ₳15,000 phone; no account, no subscription, no data plan required.
6 · Clean Water	6.1 — safe and affordable drinking water for all	Dedicated water and hygiene topics: purification methods, post-flood contamination, and the myth corrections that surround them.
9 · Infrastructure	9.1 — reliable and resilient infrastructure	The mesh is functioning communications infrastructure at exactly the moment the carrier infrastructure has failed.

All seven attributes, each with something in the repository to point at.

	Attribute	Where it appears in this project
P1	Depth of knowledge	Quantised on-device inference and KV-cache lifetime management; 768-dimension embeddings with cosine retrieval; haversine geodesy; controlled-flood routing with dedup, TTL and a hop budget; 8 kHz full-duplex audio; Android ABI targeting and R8 shrinking. A 50-query evaluation harness reporting Recall@1 and @3.
P2	Conflicting requirements	Offline capability against a 2.47 GB model. Latency against reasoning quality. Key security against serverless availability. Memory safety against per-query setup cost. Trustworthiness against helpfulness. Broadcast utility against its potential for misuse.
P3	Depth of analysis	The four-tier cascade is our own formulation. So is removing the model from the shelter path – the counter-intuitive decision that made the most urgent question 100× faster.
P4	Infrequently encountered issues	Running a 2.47 GB LLM on a mid-range handset; a GMS-free Wi-Fi Direct fallback for phones sold without Google services; Android 11 package visibility silently disabling four features until the intents were declared.
P5	Outside standards and codes	No engineering standard governs LLM-generated first aid. We built the governance: a seven-organisation source whitelist, an authoring checklist, a two-person review sign-off, corpus-only grounding, and a hard no-diagnosis rule.
P6	Diverse stakeholders	Survivors under stress and possibly one-handed; 999 operators, for whom the SOS SMS is formatted; coordinators who need aggregates; medical reviewers who need traceability; and the privacy of everyone whose location is transmitted.
P7	Interdependence	156 Dart files and eight interoperating subsystems, held together by a dependency rule that keeps the correctness core free of the framework.

The honest list. Please test us on these.

Retrieval is our weakest link

Recall@1 sits at 46%. The corpus grew from 23 chunks to 48 without moving it, which tells us the retriever — not the corpus — is the bottleneck. On-device semantic search is the current work aimed at it.

The corpus is small

48 chunks across 22 topics is a demonstration corpus, not national coverage. It is deliberately small because every chunk is manually verified against a cited source, and that review is the slow step.

The mesh is short-range

Wi-Fi Direct reaches tens of metres. Useful inside a shelter or a village cluster; not a replacement for a cellular network across a district.

The admin gate is not real authentication

It is a client-asserted check. The meaningful boundary is in the Firestore rules, and a determined attacker with the APK can get past the client side. We know, and we wrote it down.

It is Android-only, and arm64-only

The inference engine has native libraries for one architecture. No iOS build, and no emulator demo — the model genuinely will not run there.

Bangla evaluation is machine-scored

The generation test set is marked `pending_bn_review`: the gold answers have not yet been signed off by a Bangla-speaking medical reviewer. Until they are, we quote retrieval numbers and not answer-quality numbers.

Four minutes, on our own hardware, with the room's network switched off.

Time	What you will see
0:00	We put the phone into airplane mode and show you the status bar. Everything after this happens with no network of any kind.
0:20	Ask a first-aid question aloud, in Bangla. Speech recognition, retrieval, generation and speech synthesis all run on the handset.
1:30	Ask a myth as a question – "should I cut a snakebite?" – and watch the answer contradict it rather than agree.
2:10	Find the nearest cyclone shelter. Instant, because there is no model in that path.
2:40	Wi-Fi back on, cellular still off. A second phone appears as a mesh peer; we send a message and an SOS between them.
3:30	The coordinator panel on a third device shows the danger report arriving in the live list.

Why the venue's Wi-Fi cannot break this demo

The model, the corpus, the shelter database and the mesh are all on the phones we bring. There is no exhibition network dependency to fail.

Ask us anything on this list

- Turn off the network yourself, at any point, without warning us.
- Ask it something outside its scope and check that it declines instead of inventing.
- Ask where a specific answer came from – every chunk names its source document.
- Ask why a feature does *not* use AI. Several deliberately don't.
- Ask what breaks first at scale – the answer is retrieval.

What we bring

Three arm64 Android phones with the model already downloaded and permissions granted, a power bank, a screen-mirror adapter, and a recorded fallback clip we hope not to use.

One number to remember

2.47 GB of language model, answering questions in Bangla, on a phone in airplane mode, with nothing else in the room switched on.



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Guidance that stays reachable after
everything else stops.

Next: a broader reviewed corpus, a fine-tuned Bangla adapter, offline speech recognition on every device, and partnerships with the agencies whose guidance the app already cites.

948 tests passing

0 analyzer issues

Runs in airplane mode